

Radix-Packed Multi-Record Retrieval over an XPIR-Style RLWE Interface

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Abstract

Retrieving several records privately can duplicate query and evaluation work, even when single-record requests run concurrently. We study radix packing within an XPIR-style RLWE interface: one encrypted selector vector retrieves an ordered tuple, with bounded record segments combined in each reply coefficient. The construction specifies record encoding, exact plaintext capacity and digit recovery. We prove correctness under an integer no-wrap condition and derive a parametric full-task failure bound for a specified rounded Gaussian law. Single-challenge index privacy reduces to uniform-secret decisional PLWE with one additional ring sample and an explicit unit-probability loss. These claims do not certify the native sampler or concrete security. On a WSL2 host, a controlled two-record comparison against concurrent same-profile repeated XPIR covers 48 workload/profile points in cold and online views. Median paired repeated-to-packed task latency ratios range from 1.059 to 1.190; all 96 pointwise bootstrap intervals exceed one. Separate multiplicity, implementation/resource sensitivity and capacity experiments extend the finite evidence, including one two-block workload, without pooling their samples. Packing reduces native ciphertext buffers relative to repeated execution, while the flat query remains larger than the raw database at the reported lengths. The results characterize a restricted instantiation of prior weighted retrieval; the native/formal reliability bridge and transport performance remain open.

Keywords: Private information retrieval, RLWE, radix packing, XPIR, index privacy

1. Introduction

Private information retrieval (PIR) hides a client’s requested indices from the server [1, 2]. When one task needs several records, repeating a single-record protocol also repeats query generation, evaluation and extraction. Concurrent execution can overlap that work, so reducing the number of queries alone does not establish a completed-task latency benefit.

This paper studies a restricted alternative within the linear evaluation interface of XPIR [3]. The client encrypts one weighted selector vector for an ordered tuple of distinct indices. Each record is divided into bounded integer segments and encoded into polynomial coefficients. Public radix weights place the selected segments in different digit positions of each reply coefficient, allowing ordinary digit extraction after decryption. A query is one vector of N ciphertexts, not a compressed single ciphertext; the reply contains one ciphertext per polynomial block.

Encrypted weighted multi-value retrieval is established prior work [4], and radix representation is standard. The question here is how to specify its encoding, capacity and integer-noise conditions for an XPIR-style RLWE interface, and what performance this instantiation exhibits against concurrent

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same-profile repeated retrieval. Coefficient capacity, backend API support and finite functional execution are distinct constraints. In particular, a formal law cannot be inferred from a native sampler merely because observed replies decode correctly.

The paper makes three contributions:

- An explicit record-to-polynomial radix construction, with ordered recovery, exact capacity $t \geq B^\alpha$, zero padding and a native interface retaining XPIR’s linear reply arithmetic.
- Scoped formal analysis: conditional integer no-wrap correctness, a parametric full-task bound for a specified rounded Gaussian law, and single-challenge index privacy from uniform-secret decisional PLWE with $N + 1$ samples and explicit unit-probability loss.
- Separate controlled evaluations of two-record task latency, multiplicity, implementation/resource sensitivity and capacity, together with reproducible native-buffer accounting and functional checks.

The primary two-record comparison covers 48 workload/profile points in both cold and online views. Its median paired repeated-to-packed latency ratios range from 1.059 to 1.190, with all 96 pointwise intervals above one. Further experiments extend multiplicity and block coverage under their own contracts; they are not pooled with the primary evidence. These findings concern a fixed profile and host. They establish neither a global XPIR optimum nor superiority over other PIR systems, and the native/formal sampler and reliability bridge remains unresolved.

2. Related Work

2.1. XPIR and RLWE-based PIR

XPIR [3] is the closest backend reference. Its homomorphic query and database evaluation support efficient polynomial arithmetic, and its broader design includes aggregation and recursion. Our scope retains the one-dimensional linear path, with a direct SKE selector vector and explicit coefficient layout. This restricted path does not exhaust XPIR’s optimization space. The empirical conclusions below concern concurrent same-profile repeated XPIR. No measured superiority across all XPIR configurations is claimed. The pinned implementation [5] establishes artifact provenance, not a substitute citation for the XPIR paper or evidence that its native sampler realizes our formal distribution.

2.2. Multi-value and Weighted Retrieval

Hsu et al. [4, Secs. III–V] are the closest conceptual prior work. Their group-homomorphic construction already supports an encrypted weight at each selected position, encryptions of zero elsewhere, a homomorphic weighted sum, and client-side recovery of two or more values. They also discuss retrieval-count limits imposed by the plaintext modulus and a privacy argument from the underlying encryption security. Neither the multi-value goal nor that generic weighted architecture is new here.

The positional weights $K = (1, B, \dots, B^{\alpha-1})$ are public and target-independent. This instantiation replaces growing-weight remainder recovery with ordinary radix digit extraction. Radix itself is standard. The research increment is the precise connection between this choice, record-to-polynomial encoding, exact capacity and the integer no-wrap condition of the XPIR-style RLWE interface. The aggregate error depends on all encoded database polynomials, whereas direct encryption of a weight does not scale its encryption error by that weight. The formal privacy reduction and native mapping concern this instantiation; generic homomorphic privacy arguments and modulus limitations already

occur in the prior work. The conditional theorem alone establishes no concrete security-admissible region. The separately tested native API frontier in Section 6.2 concerns implementation feasibility and finite functional execution, not a native noise or security certificate.

2.3. Batch, SIMD, and Preprocessed PIR

Angel et al. [6] introduce SealPIR’s compressed queries and, separately, probabilistic batch codes (PBCs) for amortizing a batch. The latter distribute encoded database items across buckets and use a retrieval schedule, including cuckoo placement. SealPIR and the PBC transformation are separate contributions in that same paper. Vectorized BatchPIR [7] uses vectorized RLWE homomorphic encryption, bucketed batch retrieval, SIMD operations and ciphertext merging. PIRANA [8] combines constant-weight index encoding with homomorphic slot operations and a batch-code route; rotations and response organization depend on the payload and configuration. Their setup, evaluation keys and placement failure models matter to a comparison.

SimplePIR and DoublePIR are two protocols in the same paper [9]. They use LWE-based linear processing with database-dependent hints downloaded by the client; DoublePIR trades additional processing for a smaller hint. Online work cannot be compared to an entire fresh task while omitting this state and its lifecycle. These protocols provide preprocessing context rather than the direct weighted-packing precedent.

Respire [10, Secs. 2–4] directly studies small records and batch queries. It uses subrings, ring switching, query compression and response packing, and composes with PBCs for batch retrieval. Its model includes reusable client query-key material and server database preprocessing. SPIRIT [11, Secs. 2–5] combines PBC scheduling with LWE-to-RLWE conversion, shared key-expansion state and response compression. Its hintless online interface does not mean absence of server preprocessing or state. Thus neither short-record support, small-batch access, lack of an offline client hint nor response packing alone distinguishes the present work.

Radix packing here separates records by integer digit positions within each coefficient, without bucket placement or SIMD rotations to recover the targets. These architecture properties do not imply lower latency, memory or failure probability. The contribution is the explicit encoding/capacity/noise connection and its scoped evaluation within XPIR, rather than query compression, offline hints or large-batch optimization. External schemes provide mechanism and state context; their published timings are not mixed with the local measurements.

3. Preliminaries and Problem Model

3.1. Notation and Public Parameters

For a positive integer d , write $[d] = \{1, \dots, d\}$. The security parameter is κ ; PPT means probabilistic polynomial time. For an integer z and modulus $d \geq 2$, $[z]_d$ is its canonical representative in $\{0, \dots, d-1\}$. This notation extends coefficient-wise. For a monic polynomial $f \in \mathbb{Z}[X]$ of degree n , write

$$R = \mathbb{Z}[X]/(f), \quad R_q = R/qR, \quad R_t = R/tR.$$

The construction uses $f = X^n + 1$, with n a power of two, $t = 2^w$, $q > t$, and $\gcd(t, q) = 1$. The privacy reduction does not require q to be prime. All parameter descriptions have polynomial bit length in κ ; ring degrees and sample counts in the reductions are polynomially bounded. Every ring element has a unique representative of degree less than n . For $P \in R_t$, $\hat{P} \in R$ denotes the lift whose coefficients lie in $[0, t)$. This lift, followed by reduction modulo q , is an encoding into R_q , not a ring homomorphism from R_t . For $z \in R_q$, $\text{center}_q(z) \in R$ is obtained by choosing each coefficient in $[-q/2, q/2)$. The infinity norm in R is the maximum absolute value of its integer coefficients.

Table 1: Mechanism summary, without a performance ranking. State and assumptions differ across rows. The artifact provides an expanded mechanism comparison.

Scheme	Retrieval mechanism	Relation to this study
XPIR	Encrypted selection; linear evaluation; optional aggregation/recursion	Backend and empirical comparator; current path is flat.
Hsu et al.	Encrypted weights, weighted sum and integer recovery	Closest generic multi-value precedent; modulus/count limits already studied.
SealPIR / PBC	Compressed queries; separate bucketed batch transformation	Compression and placement are distinct mechanisms.
Vectorized BatchPIR	Bucket placement, SIMD processing and ciphertext merging	Batch alternative; not a local timing baseline.
PIRANA	Constant-weight selection, slots/rotations and batch codes	Alternative batch organization and payload tradeoffs.
SimplePIR / DoublePIR	LWE linear processing with client-downloaded hints	Preprocessing and amortization context.
Respire	Subring/ring-switching compression and response packing; batch codes	Direct short-record and small-batch precedent.
SPIRIT	PBCs, stateful ciphertext conversion and packed responses	Direct hintless-batch and response-packing precedent.
This construction	Several scalar digits per coefficient; unchanged linear ReplyGen	Restricted instantiation; capacity/noise/backend feasibility remains central.

The public parameters pp specify $\kappa, N, f, n, q, t, w, \ell, \rho_0, \alpha, K$, the public width $r > 0$, the family $\chi = \chi_r$ defined below, and the encoding layout; B, K, J, L are derived, with $k_\nu = B^{\nu-1}$, $\nu \in [\alpha]$. We retain $\chi_s = \chi_e = \chi_r$ as aliases in the construction and its separate correctness statement. All public parameters are fixed independently of the secret target tuple. In particular, pp is not a public encryption key. The exact radix capacity condition is $t \geq B^\alpha$, equivalently $w \geq \alpha \rho_0$, as in Lemma 1. The weights are canonical elements of $\mathbb{Z}_t$, not necessarily units; no weight coprimality or inverse is required. The condition $\gcd(t, q) = 1$ remains necessary for the security reduction.

3.2. Retrieval Task and Interfaces

A legal database is an ordered list $M = (m_1, \dots, m_N) \in (\{0, 1\}^\ell)^N$. A legal request is an ordered tuple $I = (i_1, \dots, i_\alpha) \in [N]^\alpha$ of pairwise-distinct indices; record contents at distinct indices may be equal. Write $M[I] = (m_{i_1}, \dots, m_{i_\alpha})$. The deterministic record map $\text{Encode} : \{0, 1\}^\ell \rightarrow R_t^L$ is specified in Section 4.1. We use $\text{DBEncode}(M) = (\text{Encode}(m_i))_{i \in [N]}$ only as its row-wise abbreviation, not as an additional protocol.

Only Q leaves the client; sk and st remain private. A single query is a vector of N ciphertexts, and the response contains L ciphertexts. Public parameters, including the encoding and weights, are fixed independently of the target tuple.

Table 2: Public parameters and encoding units.

Symbol	Meaning
N, n	Number of database records; coefficients per polynomial (n a power of two).
ℓ, ρ_0	Original record length and segment width, in bits; both at least one.
B, w, t	Segment bound $B = 2^{\rho_0}$; selected backend coefficient width w ; $t = 2^w$.
J, L	Segments per record $J = \lceil \ell / \rho_0 \rceil$; polynomial blocks per record $L = \lceil J / n \rceil$.
α, K	Number of distinct targets, $1 \leq \alpha \leq N$; derived radix weights $K = (1, B, \dots, B^{\alpha-1})$.
ℓ_{enc}	Full-block encoded bit length Lnw ; distinct from ℓ and Jw .
χ_r, r	Coefficient-wise rounded Gaussian family and its public continuous standard deviation $r > 0$; $\chi_s = \chi_e = \chi_r$.
p_{unit}	Probability that a uniform element of R_q is invertible.

Table 3: Protocol interfaces and ownership. Public parameters determine all dimensions. Query generation is randomized; reply generation and extraction are deterministic for their inputs.

Algorithm	Party	Input and output
KeyGen	Client	$pp \mapsto sk = s$; one fresh secret per query.
QueryGen	Client	$(pp, sk, I) \mapsto (Q, \text{st})$, where $Q \in (R_q^2)^N$ and private $\text{st} = I$.
ReplyGen	Server	$(pp, P, Q) \mapsto \text{Resp} \in (R_q^2)^L$, where $P = \text{DBEncode}(M) \in R_t^{N \times L}$.
ReplyExt	Client	$(pp, sk, \text{Resp}, \text{st}) \mapsto \widehat{M} \in (\{0, 1\}^\ell)^\alpha$, in the requested order.

3.3. Correctness Goal

For every admissible pp , legal M , and legal I , consider the honest execution

$$sk \leftarrow \text{KeyGen}(pp), \quad (Q, \text{st}) \leftarrow \text{QueryGen}(pp, sk, I),$$

$$\text{Resp} = \text{ReplyGen}(pp, \text{DBEncode}(M), Q), \quad \widehat{M} = \text{ReplyExt}(pp, sk, \text{Resp}, \text{st}).$$

Define $\text{Corr} = [\widehat{M} = M[I]]$. A correctness target $\Pr[\neg \text{Corr}] \leq \delta_{\text{corr}}(pp)$ quantifies over all such databases and requests, with probability over key generation and all encryption randomness. This target is a specification, not an unconditional negligible-failure claim. Section 5.1 proves ordered recovery under an integer no-wrap condition and derives a bound for the specified error law.

3.4. Single-challenge Index Privacy

Definition 3.1 (Index privacy for packed multi-record retrieval). Fix admissible public parameters pp independently of the targets. The database and target tuples must be legal as defined above; both challenge tuples use the same public layout and weight tuple. For a two-stage PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, the experiment is:

1. The challenger samples $sk \leftarrow \text{KeyGen}(pp)$.
2. $\mathcal{A}_1(pp)$ outputs (M, I_0, I_1, σ) , where σ is its polynomial-length private state. Both tuples and the database must be legal. The tuples may coincide, covering degenerate instances such as $N = \alpha = 1$. On illegal input, the experiment terminates with an independent fair success bit.

3. The challenger samples $b \leftarrow \{0, 1\}$, computes

$$(Q, \text{st}) \leftarrow \text{QueryGen}(pp, sk, I_b), \quad P = \text{DBEncode}(M),$$

$$\text{Resp} = \text{ReplyGen}(pp, P, Q),$$

and gives (Q, Resp) to the adversary.

4. $b' \leftarrow \mathcal{A}_2(pp, M, I_0, I_1, \sigma, Q, \text{Resp})$. The experiment succeeds iff $b' = b$.

Neither sk nor the client state st is disclosed. Define

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{priv}}(\kappa) = \left| \Pr[\text{experiment succeeds}] - \frac{1}{2} \right|.$$

Single-challenge index privacy means this advantage is negligible for every PPT adversary. The probability includes key generation, all encryptions, the challenge bit, and adversary randomness.

The public parameters, database, and adversary state are part of the server's view. Its deterministic intermediate computations are functions of these values and the query. No decryption outcome or client extraction state is included in this game.

3.5. Specified Error Law

We adopt the coefficient-wise Gaussian family explicitly defined by Brakerski and Vaikuntanathan [12, Sec. 1.3]:

$$Y_j \leftarrow \mathcal{N}(0, r^2), \quad Z_j = \text{round}(Y_j), \quad \chi_r = \mathcal{L}\left(\sum_{j=0}^{n-1} Z_j X^j\right),$$

where the Y_j are independent and rounding is to the nearest integer. Equivalently, one coefficient has probability mass

$$p_r(z) = \int_{z-1/2}^{z+1/2} \frac{\exp(-y^2/(2r^2))}{\sqrt{2\pi} r} dy, \quad z \in \mathbb{Z}.$$

Half-integer ties have probability zero under the ideal continuous law. Both the secret and every fresh encryption error use this same product distribution, independently across coefficients and draws. Here r is the standard deviation before rounding; its numerical value remains to be selected jointly with the other parameters.

This rounded law differs from the integer Gaussian with mass proportional to $\exp(-\pi z^2/\eta^2)$ and is not identified here with a canonical-embedding Gaussian. Efficient machine realization and its approximation to this ideal law require separate justification.

3.6. Linear Encryption Interface

We use the symmetric-key algebraic interface of the Ring-LWE construction of Brakerski and Vaikuntanathan [12], in the linear evaluation setting of XPIR [3]. All ciphertext arithmetic below is in R_q ; all message lifts use canonical coefficients.

$$\begin{aligned} \text{KeyGen}(pp) : & \quad s \leftarrow \chi, \quad sk = s, \\ \text{Enc}_s(v) : & \quad a \leftarrow R_q \text{ uniformly}, \quad e \leftarrow \chi, \quad C = (as + te + \hat{v}, -a), \\ \text{Dec}_s(C) : & \quad [\text{center}_q(c_0 + c_1 s)]_t, \\ \text{Add}(C, C') : & \quad (c_0 + c'_0, c_1 + c'_1), \\ \text{Absorb}(P, C) : & \quad (\hat{P}c_0, \hat{P}c_1). \end{aligned}$$

Each encryption samples fresh independent a, e , including encryptions of zero. The client alone possesses sk , encrypts queries, and decrypts replies. The server needs only pp , the encoded database, and ciphertexts to perform **Add** and **Absorb**. No ciphertext-ciphertext multiplication or public key is required.

XPIR's mathematical representation (a, b) , with $b = as + te + v$, corresponds to our $C = (b, -a)$, so $c_0 + c_1 s = b - as$. This correspondence makes no claim of identical binary serialization. The final mod- t step returns canonical coefficients, whereas the preceding mod- q step uses centered coefficients. Correctness requires control of the integer value before reduction modulo q , as formalized in Theorem 1.

3.7. Uniform-secret Decisional PLWE

Definition 3.2 (Uniform-secret decisional PLWE). Fix a publicly specified parameter family (f, q, χ) , where χ is efficiently sampleable over R . For a polynomially bounded $\tau = \tau(\kappa)$, consider

$$\mathcal{U}_{\text{real}}^{(\tau)} = ((a_i, a_i s + e_i))_{i=1}^\tau, \quad \mathcal{U}_{\text{unif}}^{(\tau)} = ((a_i, u_i))_{i=1}^\tau$$

in $(R_q^2)^\tau$. One secret $s \leftarrow U(R_q)$ is shared by all samples. All $a_i, u_i \leftarrow U(R_q)$ and $e_i \leftarrow \chi$ are mutually independent and independent of s ; errors are reduced modulo q when used in R_q . For a PPT distinguisher $\mathcal{B}$, define

$$\text{Adv}_{\mathcal{B}}^{\text{uPLWE}}(\kappa; \tau) = \left| \Pr[\mathcal{B}(pp, \mathcal{U}_{\text{real}}^{(\tau)}) = 1] - \Pr[\mathcal{B}(pp, \mathcal{U}_{\text{unif}}^{(\tau)}) = 1] \right|.$$

The τ -sample assumption for this family is that this advantage is negligible in κ for every PPT $\mathcal{B}$.

This is a distribution-specific decisional assumption, following the uniform-secret PLWE formulation discussed in [12, Sec. 2]. Hardness is assumed only for the corresponding instance, not for every efficiently sampleable χ . We make no worst-case ideal-lattice reduction claim and certify no concrete security level.

4. Radix-Packed Retrieval

The construction represents segments from different records as digits inside a coefficient. It uses direct encryptions of public radix weights and retains the linear reply interface of Section 3.6. The encoding, selector, and extraction must share the same public layout.

4.1. Record Encoding and Decoding

Fix an ordered record $m_i = (b_{i,0}, \dots, b_{i,\ell-1}) \in \{0, 1\}^\ell$. Set $b_{i,v} = 0$ for $v \geq \ell$. For $h \in [J]$, define

$$x_{i,h} = \sum_{v=0}^{\rho_0-1} 2^v b_{i,(h-1)\rho_0+v}, \quad 0 \leq x_{i,h} < B = 2^{\rho_0}. \quad (1)$$

Set $x_{i,h} = 0$ for $h > J$. Thus an incomplete final segment is padded with zero high bits. Define $\text{Encode}(m_i) = (P_{i,1}, \dots, P_{i,L})$, where

$$P_{i,j}(X) = \sum_{u=0}^{n-1} x_{i,(j-1)n+u+1} X^u \in R_t, \quad j \in [L]. \quad (2)$$

The indices are $i \in [N]$, $j \in [L]$, $h \in [J]$, and $u \in \{0, \dots, n-1\}$; their relation is $h = (j-1)n + u + 1$. Each coefficient holds one scalar segment, and each polynomial holds up to n segments. The final polynomial is padded with zero coefficients.

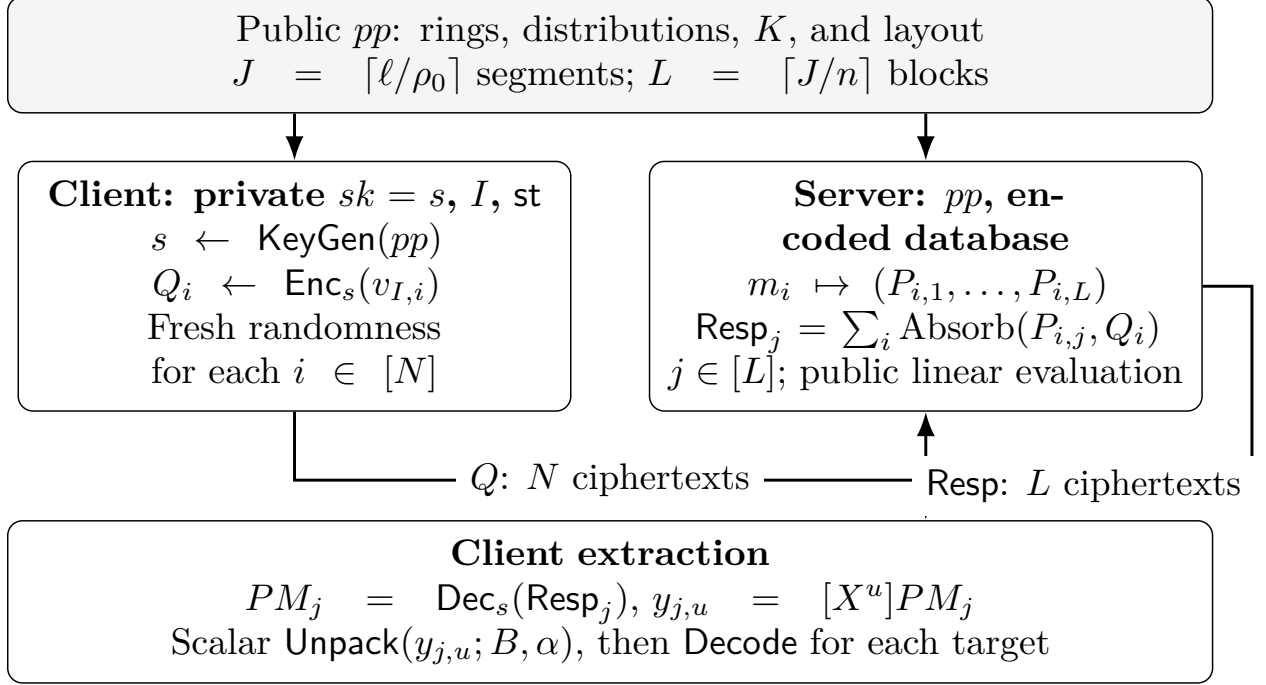


Figure 1: SKE query-reply-extraction flow. Secret-key generation and client state are local to the client. Each reply encrypts one packed polynomial PM_j , containing up to n packed scalar coefficients; decryption and extraction are correct under Theorem 1.

Decode reads canonical coefficients in increasing j , then increasing u , keeps the first J segments, and expands each into ρ_0 bits in increasing bit position. It concatenates these bits and retains the first ℓ . For the image of **Encode**, uniqueness of the binary expansion in $[0, B)$ recovers every bit in (1); the discarded bits and coefficients are precisely the padding. Consequently

$$\text{Decode}(\text{Encode}(m_i)) = m_i.$$

4.2. Packed Reply Structure

For target ordinal $\nu \in [\alpha]$, associate i_ν with $k_\nu = B^{\nu-1}$. The intended plaintext of reply block j is

$$PM_j = \sum_{\nu=1}^{\alpha} k_\nu P_{i_\nu, j} \in R_t. \quad (3)$$

The client decrypts each block, extracts the canonical radix digits, and decodes the resulting record polynomials (Fig. 1).

4.3. Scalar Packing and Unpacking

Lemma 1 (Radix packing/unpacking). *Let $B = 2^{\rho_0}$, $\rho_0 \geq 1$, $\alpha \geq 1$, and $x_\nu \in \{0, \dots, B-1\}$. Set*

$$K = (1, B, \dots, B^{\alpha-1}), \quad k_\nu = B^{\nu-1}. \quad (4)$$

Then $y = \sum_{\nu=1}^{\alpha} B^{\nu-1} x_\nu$ satisfies $0 \leq y < B^\alpha$, has a unique base- B representation, and $\text{Unpack}(y; B, \alpha)$ returns its digits by

$$x_\nu = \left\lfloor \frac{y}{B^{\nu-1}} \right\rfloor \bmod B, \quad \nu \in [\alpha]. \quad (5)$$

Algorithm 1 Query Generation $\text{QueryGen}(pp, sk, I)$

Require: Admissible pp , client key $sk = s$, legal ordered target tuple I .

Ensure: Query Q , private client state st .

- 1: **for** $i = 1$ to N **do**
 - 2: Determine $v_{I,i}$ by (6).
 - 3: $Q_i \leftarrow \text{Enc}_s(v_{I,i})$ using fresh independent a_i, e_i .
 - 4: **end for**
 - 5: $\text{st} \leftarrow I$.
 - 6: **return** $((Q_1, \dots, Q_N), \text{st})$.
-

All such integers embed without reduction in $\mathbb{Z}_t$ exactly when $t \geq B^\alpha$. For $t = 2^w$, this is equivalent to $w \geq \alpha \rho_0$.

Proof. The geometric sum gives

$$0 \leq y \leq (B - 1) \sum_{\nu=1}^{\alpha} B^{\nu-1} = B^\alpha - 1.$$

For each ν , the contribution from lower positions is less than $B^{\nu-1}$, while each higher position is a multiple of B^ν . Integer division by $B^{\nu-1}$, followed by reduction modulo B , therefore returns exactly x_ν . This deterministic inverse proves uniqueness. Conversely, every integer in $[0, B^\alpha)$ has these α digits, so the domain has exactly B^α values. Its canonical embedding into $[0, t)$ is possible exactly when $t \geq B^\alpha$, including equality. When $\alpha = 1$, $K = (1)$ and the decoder returns $x_1 = y$. $\square$

The weights need not be units modulo t . Positional digit recovery therefore needs no coprimality condition on individual weights, unlike growing-weight remainder recovery [4].

4.4. Query Generation

Define the constant polynomial

$$v_{I,i} = \begin{cases} B^{\nu-1}, & i = i_\nu \text{ for some } \nu \in [\alpha], \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

It is a constant polynomial, not a polynomial with every coefficient equal to k_ν . Algorithm 1 calls encryption exactly once per coordinate. It neither reuses one encryption of zero nor replaces direct $\text{Enc}_s(k_\nu)$ by $k_\nu \text{Enc}_s(1)$.

4.5. Reply Generation

Algorithm 2 uses the encoded polynomials. One ciphertext is returned per polynomial block, regardless of whether its final coefficients are padding. Only the client decrypts the response.

4.6. Reply Extraction

Algorithm 3 applies Lemma 1 to each integer coefficient $y_{j,u}$, then uses the record decoding already defined in Section 4.1. Public pp fixes all lengths, and derived radix weights. With $\alpha = 1$, $K = (1)$, the construction reduces to the single-record reference flow in Section 4.6.

Single-record reference.. With $\alpha = 1$ and $K = (1)$, the query independently encrypts one at the target and zero elsewhere. The reply contains L ciphertexts, one for each record polynomial block. This is the flat XPIR-style reference flow; native layouts outside this interface can have different block counts.

Algorithm 2 Reply Generation $\text{ReplyGen}(pp, \{P_{i,j}\}, Q)$

Require: Public pp , encoded database $P_{i,j}$ for $i \in [N], j \in [L]$, query Q .

Ensure: Exactly L reply ciphertexts.

- 1: **for** $j = 1$ to L **do**
 - 2: $\text{Resp}_j \leftarrow \sum_{i=1}^N \text{Absorb}(P_{i,j}, Q_i)$, using **Add**.
 - 3: **end for**
 - 4: **return** $(\text{Resp}_1, \dots, \text{Resp}_L)$.
-

Algorithm 3 Reply Extraction $\text{ReplyExt}(pp, sk, \text{Resp}, \text{st})$

Require: pp , client key $sk = s$, L reply ciphertexts, private $\text{st} = I$.

Ensure: Recovered tuple in the order of I , under Theorem 1.

- 1: **for** $j = 1$ to L **do**
 - 2: $PM_j \leftarrow \text{Dec}_s(\text{Resp}_j)$.
 - 3: **for** $u = 0$ to $n - 1$ **do**
 - 4: $y_{j,u} \leftarrow [X^u]PM_j$, as an integer in $[0, t)$.
 - 5: $(\hat{x}_{1,h}, \dots, \hat{x}_{\alpha,h}) \leftarrow \text{Unpack}(y_{j,u}; B, \alpha)$, where $h = (j - 1)n + u + 1$.
 - 6: **end for**
 - 7: **end for**
 - 8: **for** $\nu = 1$ to α **do**
 - 9: Form $\hat{P}_{\nu,j} = \sum_{u=0}^{n-1} \hat{x}_{\nu,(j-1)n+u+1} X^u$ for $j \in [L]$.
 - 10: $\hat{m}_{i_\nu} \leftarrow \text{Decode}(\hat{P}_{\nu,1}, \dots, \hat{P}_{\nu,L})$.
 - 11: **end for**
 - 12: **return** $(\hat{m}_{i_1}, \dots, \hat{m}_{i_\alpha})$.
-

Native representation.. An implementation must zero-extend each ρ_0 -bit segment into its declared w -bit field, pad to Ln fields, and import exactly the polynomials in (2). Extraction uses the same layout and trims to ℓ bits. Width changes require rebuilding preprocessing; weights must be exactly representable by the encryption interface. These representation conditions preserve the mathematics but do not guarantee native support for every admissible parameter choice.

5. Correctness and Index Privacy

5.1. Conditional Correctness

Let $e_i \in R$ be the actual independently sampled encryption errors. Before any coefficient reduction modulo q , define in the integer ring R

$$\widehat{PM}_j = \sum_{\nu=1}^{\alpha} B^{\nu-1} \hat{P}_{\nu,j}, \quad \hat{E}_j^{\text{agg}} = \sum_{i=1}^N \hat{P}_{i,j} e_i, \quad Z_j = \widehat{PM}_j + t \hat{E}_j^{\text{agg}}. \quad (7)$$

These products reduce only by $X^n + 1$. Under the radix capacity condition, $\widehat{PM}_j$ is also the canonical lift of PM_j : every coefficient lies in $[0, B^\alpha)$, hence $\|\widehat{PM}_j\|_\infty \leq B^\alpha - 1$. Noise products are negacyclic convolutions and can have negative integer coefficients.

Theorem 1 (Conditional correctness). *Fix admissible pp , a legal database and target tuple, and any realization of key generation and query encryption. Assume that for every $j \in [L]$,*

$$\|\widehat{PM}_j\|_\infty + t \|\hat{E}_j^{\text{agg}}\|_\infty < q/2. \quad (8)$$

A sufficient, data-independent plaintext bound is

$$B^\alpha - 1 + t\|\widehat{E}_j^{\text{agg}}\|_\infty < q/2 \quad (j \in [L]).$$

Under (8), Algorithms 1–3 return $(m_{i_1}, \dots, m_{i_\alpha})$.

Proof. For each query ciphertext, $c_{0,i} + c_{1,i}s \equiv v_{I,i} + te_i \pmod{qR}$. If $\text{Resp}_j = (d_{0,j}, d_{1,j})$, the ring arithmetic in Algorithm 2 gives

$$\begin{aligned} d_{0,j} + d_{1,j}s &\equiv \sum_i \widehat{P}_{i,j}(v_{I,i} + te_i) \pmod{qR} \\ &\equiv \widehat{PM}_j + t\widehat{E}_j^{\text{agg}} = Z_j \pmod{qR}. \end{aligned}$$

This is a congruence; it does not identify the integer linear combination of canonical ciphertext representatives with Z_j . By (8), $\|Z_j\|_\infty < q/2$. Thus $\text{center}_q(d_{0,j} + d_{1,j}s) = Z_j$ in R , and

$$\text{Dec}_s(\text{Resp}_j) = [Z_j]_t = [\widehat{PM}_j]_t = PM_j.$$

For each u , its canonical coefficient is exactly $y_{j,u} = \sum_\nu k_\nu x_{i_\nu, (j-1)n+u+1}$, without plaintext wraparound. Lemma 1 recovers all target segments, including zero padding. The Encode/Decode identity then recovers every target record. $\square$

Direct encryption of k_ν does not multiply its error e_i by k_ν . The weights increase the plaintext term; the data polynomials determine noise accumulation through $\sum_i \widehat{P}_{i,j}e_i$, including errors from non-target coordinates. Bounding a value only after centering would not rule out prior wraparound, and is not the argument above.

5.2. Parametric Full-Task Correctness

We now bound the failure of one complete ordered α -record retrieval under the formal χ_r , for any $r > 0$. Theorem 1 remains the deterministic implication used below; no numerical r or cryptanalytic security estimate is needed for this analysis.

Lemma 2 (Rounded-noise tails). *Let $Y \sim \mathcal{N}(0, r^2)$ and $Z = \text{Round}(Y)$. For every integer $h \geq 0$,*

$$p_h(r) := \Pr(|Z| > h) = \text{erfc}\left(\frac{h + 1/2}{\sqrt{2}r}\right). \quad (9)$$

For integer $h \geq 1$, $\Pr(|Z| \geq h) = p_{h-1}(r)$, with threshold $(h - 1/2)/(\sqrt{2}r)$ in the complementary error function.

The proof is given in Appendix Appendix A.1.

Write $e_i = \sum_{u=0}^{n-1} e_{i,u}X^u$ and define the single event

$$G_h = \{ |e_{i,u}| \leq h \text{ for every } i \in [N], 0 \leq u < n \}. \quad (10)$$

Coefficient-wise and draw-wise independence give

$$\Pr(G_h^c) = 1 - (1 - p_h(r))^{Nn} \leq Nn p_h(r). \quad (11)$$

All N query errors, including non-target coordinates, enter this event. The same errors affect every reply block; neither L nor α introduces an additional union factor.

Lemma 3 (Block support and negacyclic noise). *For $j \in [L]$, let*

$$m_j = \min\{n, \max\{0, J - (j - 1)n\}\}.$$

Then $1 \leq m_j \leq n$ and $\|\widehat{P}_{i,j}\|_1 \leq m_j(B - 1)$. For any signed $P, e \in \mathbb{Z}[X]/(X^n + 1)$, $\|Pe\|_\infty \leq \|P\|_1\|e\|_\infty$. Consequently, on G_h , simultaneously for all j ,

$$\|\widehat{E}_j^{\text{agg}}\|_\infty \leq Nm_j(B - 1)h. \quad (12)$$

The proof is given in Appendix Appendix A.2.

Proposition 1 (Full-task correctness under rounded Gaussian errors). *Fix admissible pp , a legal database M , and a legal ordered tuple $I = (i_1, \dots, i_\alpha)$. Use the formal iid law χ_r , $r > 0$, and generate a fresh key, query and reply by the stated algorithms. If integer $h \geq 0$ satisfies*

$$B^\alpha - 1 + tNm_j(B - 1)h < q/2 \quad (j \in [L]), \quad (13)$$

then the probability that $\text{ReplyExt}(pp, sk, \text{Resp}, st)$ fails to return $(m_{i_1}, \dots, m_{i_\alpha})$, in that order, is at most

$$1 - (1 - p_h(r))^{Nn}.$$

The probability includes KeyGen, all query-encryption errors, all a_i , and all other formal encryption randomness.

Proof. On G_h , Lemma 3 and $\|\widehat{PM}_j\|_\infty \leq B^\alpha - 1$ make (13) imply (8) for every block. Theorem 1 then recovers every packed block and every requested record in the prescribed order. Task failure is thus contained in G_h^c , whose probability is (11). This implication holds for every realization of the secret and the uniform a_i ; averaging over them adds no failure term. $\square$

Exact integer threshold.. Set $A = B^\alpha - 1$, $D_j = tNm_j(B - 1) > 0$, and $C = (q - 1)/2 - A$. Since $t = 2^w$ and $\gcd(t, q) = 1$ make q odd,

$$A + D_j h < q/2 \iff A + D_j h \leq (q - 1)/2.$$

If $C < 0$, no nonnegative integer h passes this sufficient screen; this does not prove impossibility. Otherwise define

$$h_{\max,j} = \left\lfloor \frac{C}{D_j} \right\rfloor, \quad h_{\max} = \min_{j \in [L]} h_{\max,j}. \quad (14)$$

Then $h_{\max}$ passes every block and $h_{\max} + 1$ fails at least one limiting block. Because $p_h(r)$ decreases with h , this choice gives the strongest bound within this uniform coefficient-event method. Define the resulting ideal-law failure upper bound

$$\epsilon_{\text{dec}}(r) = 1 - \left(1 - \text{erfc}\left(\frac{h_{\max} + 1/2}{\sqrt{2}r}\right)\right)^{Nn}. \quad (15)$$

It is the exact probability of $G_{h_{\max}}^c$, not necessarily the actual decryption-failure probability.

Corollary 1 (Union bound and a sufficient noise region). *Under the proposition's hypotheses, $\Pr[\text{task failure}] \leq Nn p_h(r)$. When $C \geq 0$, for any ideal target $0 < \varepsilon < 1$, the region*

$$0 < r \leq \frac{h_{\max} + 1/2}{\sqrt{2} \text{erfc}^{-1}(\varepsilon/(Nn))} \quad (16)$$

suffices for $\epsilon_{\text{dec}}(r) \leq \varepsilon$.

Proof. Use (11). On $(0, 1)$, erfc^{-1} is positive and decreasing, so (16) is equivalent to $Nn p_{h_{\max}}(r) \leq \varepsilon$. $\square$

The exact-product bound also gives a symbolic endpoint: put $\delta_* = 1 - (1 - \varepsilon)^{1/(Nn)}$ and replace $\varepsilon/(Nn)$ by δ_* in (16). This is equivalent to $\epsilon_{\text{dec}}(r) \leq \varepsilon$ for the bound function and is at least as permissive as the union region. No active-profile numerical inverse or noise value is selected here.

5.3. Normal Form and Ciphertext Pseudorandomness

Let $\mathcal{F}_{\text{real}}^{(\tau)} = ((A_i, A_i s + E_i))_{i=1}^\tau$ have $s \leftarrow \chi$, independent $E_i \leftarrow \chi$, and independent uniform $A_i \in R_q$. Let $\mathcal{F}_{\text{unif}}^{(\tau)}$ be uniform in $(R_q^2)^\tau$, and define $\text{Adv}_{\mathcal{D}}^{\text{NF}}(\kappa; \tau)$ as the corresponding acceptance-probability gap. This is an intermediate distribution, not an additional hardness assumption. Write

$$p_{\text{unit}} := \Pr_{a \leftarrow U(R_q)}[a \in R_q^\times].$$

The normal-form method is established in the discussion preceding Definition 1 of [12] and in [13, Lemma 2.24 and Claim 2.25]. The following discrete-polynomial specialization records its sample count and loss in our notation; it is not a new reduction.

Corollary 2 (Normal-form specialization). *For every PPT $\mathcal{D}$ on τ normal-form samples, there is a PPT $\mathcal{B}$ on $\tau + 1$ uniform-secret samples such that*

$$\text{Adv}_{\mathcal{D}}^{\text{NF}}(\kappa; \tau) \leq \frac{\text{Adv}_{\mathcal{B}}^{\text{uPLWE}}(\kappa; \tau + 1)}{p_{\text{unit}}}.$$

See Appendix Appendix A.3 for the specialization proof.

This consumes one additional *ring* sample. Lemma 2.24 of [13] is stated with continuous errors and a discretized secret; it does not automatically make both laws the same discrete χ . The transformation in Appendix Appendix A.3 verifies that property for our already-discrete input samples. For cyclotomic R , Claim 2.25 of [13] supplies $p_{\text{unit}} \geq 1/\text{poly}(n, \log q)$ for any integer $q \geq 2$. For the special case of prime $q \equiv 1 \pmod{2n}$ and $f = X^n + 1$, it gives $p_{\text{unit}} = (1 - 1/q)^n \geq e^{-1}$. We use the general lower bound for the construction; prime q is not a theorem requirement. For a general monic f , the quantitative bound still retains p_{unit} , without asserting this cyclotomic lower bound.

Corollary 3 (XPIR-SKE ciphertext pseudorandomness). *If $\gcd(t, q) = 1$, pseudorandomness of the normal-form joint distribution implies XPIR-SKE ciphertext pseudorandomness for any fixed message vector in R_t^t . The transformation preserves the distinguisher's advantage and sample count.*

The bijective scaling argument is in Appendix Appendix A.4.

The message vector may also be fixed conditional on public, sample-independent side information, as in Definition 3.1.

5.4. Reduction from Uniform-secret PLWE

Theorem 2 (Single-challenge index privacy of the formal construction). *Fix the formal construction with $f = X^n + 1$, n a power of two, a specified efficiently sampleable χ , and $\gcd(t, q) = 1$. For every PPT adversary $\mathcal{A}$ in Definition 3.1, there is a PPT $\mathcal{B}$ using $N + 1$ uniform-secret PLWE samples such that*

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{priv}}(\kappa) \leq \frac{\text{Adv}_{\mathcal{B}}^{\text{uPLWE}}(\kappa; N + 1)}{p_{\text{unit}}}.$$

Here $N, n, \log q$, the public parameter descriptions, and the encoded database length are polynomially bounded in κ . Consequently, the corresponding $(N + 1)$ -sample uniform-secret assumption in Definition 3.2 implies single-challenge index privacy, since p_{unit} has the inverse-polynomial lower bound stated in Section 5.3.

Proof outline. The normal-form reduction provides N shared-secret pairs from $N + 1$ uniform-secret samples. Scaling by the unit t and translating by the selected constants simulates the entire query jointly, with no per-target hybrid. A uniform query and its deterministic reply are independent of the challenge bit. The normal-form abort branch multiplies both acceptance probabilities by p_{unit} , giving the stated loss under the two declared advantage conventions. Appendix Appendix A.5 gives the complete simulation, including illegal inputs, the unconditional uniform acceptance probability, and polynomial-time conditions. $\square$

The dependency chain is uniform-secret PLWE with $N + 1$ samples, the existing normal-form reduction to N shared-secret samples, the SKE scaling and message-translation step, and finally the reduction to packed index privacy above. The extra ring sample is a reduction input, not an additional ciphertext sent by the client.

Scope.. The proof establishes only the defined single-challenge index privacy under the corresponding uniform-secret assumption for (f, q, χ) . It makes no worst-case ideal-lattice reduction claim and does not require prime q . It does not establish adaptive multi-round security, CCA security, SPIR, malicious correctness, or a QROM claim. Unpack, Decode, and all client decryption results are absent from the adversary’s view. Correctness, no-wrap, and decryption-failure probability therefore play no role in this privacy proof. Theorem 1 remains a separate conditional correctness result.

Key lifetime.. Each PIR query uses a fresh secret key, authorized only for that query’s N SKE ciphertexts. Completion, abort, or retry retires its encryption authority; a retry or next query runs KeyGen again. No additional diagnostic or public encryptions under that key are permitted. Thus a completed query has $\tau_{\text{visible}} = N$ ciphertexts per key (an aborted query has at most N); Theorem 2 still uses $N + 1$ uPLWE input samples. Retaining the key to decrypt its outstanding reply does not authorize further encryption. This policy introduces no key-reuse or multi-query security claim.

6. Performance Evaluation

6.1. Implementation and Measurement Contract

The native implementation retains XPIR’s linear reply evaluation and NTT/CRT arithmetic [3, 5]. It fixes polynomial degree 4096, recursion dimension one, aggregation factor one and a 122-bit ciphertext modulus (the product of two backend RNS primes). The layout uses $w = \alpha\rho_0$, so $t = B^\alpha$. Exact build identities and parameters are provided with the artifact. The native bounded, biased sampler and OS-seeded Salsa20 generator are unchanged; they are not identified with the formal rounded Gaussian law. The implementation evidence therefore concerns execution and performance, with the security boundary discussed in Section 7.

All experiments use a Ryzen 7 3700X running Ubuntu under WSL2. Frequency, turbo and host scheduling are host-managed. Each packed worker and each repeated worker has a fresh client key, private client state and private database preprocessing. The comparator runs α same-profile single-record XPIR requests concurrently, with no artificial additional network round trip. Methods share raw records, ordered targets and public layout within each comparison. This measures complete tasks under a specified lifecycle; it does not isolate radix arithmetic from setup, worker or preprocessing costs.

Table 4: Separate experimental designs. A cell fixes workload, profile, view and, where applicable, implementation/resource bundle.

Experiment	Cells	Measured pairs	Design per cell
Primary two-record	96	2160	3 warmups; 30, 30, 20 or 10 pairs as N increases.
Multiplicity	24	240	3 warmups; 10 pairs.
Bundle sensitivity	8	240	3 successive sessions, each with 3 warmups and 10 pairs.
Capacity	10	300	Same session design; five record lengths in both views.

Timing and estimands. Task latency extends from client configuration through reply extraction, using monotonic raw-clock endpoints. A repeated task’s wall time is the latest worker end minus the earliest worker start. The *cold* view includes fresh server configuration, database encoding and native import; the *online* view stages this private database state before timing. Both views include fresh client setup/key generation, query generation, reply generation and extraction, including buffer copies in those phases. Process startup, fixture generation, logging, teardown and validation are excluded; validation begins after all peer endpoints. Online results do not model amortized key reuse.

For each cell, the main estimator is the median of paired ratios $R = T_{\text{repeated}}/T_{\text{packed}}$, distinct from the ratio of separate method medians. The primary and multiplicity experiments use 10,000 paired-bootstrap resamples per cell and pointwise 95% percentile intervals, with fixed seeds reset per cell. Their interleaved, balanced schedules do not establish independent sessions or justify coverage under temporal dependence. Intervals are not simultaneous across cells. Warmups are excluded; failures are retained without retries, deletion or imputation. All reported measured pairs are complete.

6.2. Native Feasibility

Structural capacity, API support and finite encrypted execution are separate checks. The nine tested combinations of $\alpha \in \{2, 3, 4\}$ and $\rho_0 \in \{8, 12, 16\}$ all meet exact radix capacity and the zero-noise arithmetic screen. Six are supported by the current API: all three two-record profiles, (3, 8), (3, 12) and (4, 8). The other three exceed the direct integer weight limit $2^{32} - 1$; (4, 16) also exceeds the checked width range $1 \leq w \leq 56$. These are implementation limits, not mathematical impossibility results. The supplemental frontier table lists every profile.

The two-record profiles each pass 82 encrypted cases; the three supported additional profiles each pass ten. Coverage includes ordered recovery, distinct indices with equal contents, maximum digits and partial-segment padding. A separate capacity-focused suite passes 208 tasks with 448 fresh-key workers, including zero, maximum-segment, alternating and deterministic pseudorandom fixtures, reversed targets and full-block checks. Small databases at 32767, 32768 and 32769 bits cross the $J = n$ boundary; these are functional checks, not additional performance samples. Exact integer checks retain each block’s admissible $h_{\max, j}$ and failing successor. None of these finite checks calibrates a native failure rate.

6.3. Primary Two-Record Comparison

This experiment fixes $\alpha = 2$, with $N \in \{256, 1024, 4096, 16384\}$, $\ell \in \{256, 512, 1024, 2048\}$ bits and $\rho_0 \in \{8, 12, 16\}$. All workloads have $L = 1$. Packed retrieval uses one arithmetic thread with affinity to two logical CPUs; the two repeated workers are pinned separately to those CPUs. The parent has the same two-CPU allowance. The packed address-space cap is 16 GiB; each repeated worker has an 8 GiB cap. The aggregate sampled RSS cap is 16 GiB. This allowance does not make packed arithmetic parallel.

Table 5: Absolute task latency at the smallest and largest primary workload corners, with $\rho_0 = 8$. These four cells are descriptive examples; Fig. 2 retains the full grid.

N	ℓ (bits)	View	Packed (s)	Repeated (s)	Paired R
256	256	cold	0.206	0.223	1.099
256	256	online	0.169	0.180	1.069
16384	2048	cold	14.315	16.743	1.164
16384	2048	online	12.036	13.509	1.146

Time columns are separate method medians. Their quotient is not the median paired ratio in the last column.

Packed retrieval has lower separate median task latency in every tested cell. The 96 median paired ratios range from 1.059 to 1.190, and all pointwise lower interval endpoints exceed one. Figure 2 retains the complete grid; Table 5 gives absolute times at two workload corners. The range describes cell medians, not an overall speedup or a family-wise inference. It compares this flat, fixed-profile XPIR path, without establishing superiority over optimized XPIR profiles.

6.4. Multiplicity Experiment

The multiplicity experiment varies $\alpha \in \{2, 3, 4\}$ at $N \in \{1024, 4096\}$, $\ell \in \{512, 2048\}$, $\rho_0 = 8$ and $w = 8\alpha$, again with $L = 1$. Packed execution is a serial worker pinned to one logical CPU. Repeated execution uses α private workers on distinct logical CPUs from a four-CPU set, which also bounds the parent. Each worker has an 8 GiB address-space limit; aggregate virtual memory has a separately sampled 16 GiB limit. Both width and repeated-worker CPU count change with multiplicity. This is not a fixed-total-core scaling comparison, and visible WSL core IDs do not establish physical host isolation.

All 24 observed median ratios exceed one. Pointwise intervals exclude one in all eight cells at each of $\alpha = 3, 4$, but only three of eight at $\alpha = 2$. All eight fixed-workload/view rows have $R_2 < R_3 < R_4$ (Table 6); this ordering is descriptive, without a monotonicity significance test or universal scaling claim.

The two-record entries are internal references for this experiment. All eight are smaller than their matched primary-experiment ratios, with disjoint experiment-specific intervals. Extraction scope, API wrappers, affinity, parent allowance, fixtures and sample schedules differ. In particular, the primary adapter reads record-used fields, while the multiplicity adapter extracts complete polynomial blocks with checked wrappers. These differences have no established causal attribution. The experiments remain separate; neither estimate is selected as the true effect, and their samples are not pooled.

6.5. Bundle Sensitivity and Capacity

The sensitivity experiment crosses two extraction/API bundles (record-field and full-block) with the primary two-CPU and multiplicity four-CPU-parent resource allowances. It holds $N = 1024$, $\ell = 512$, $\alpha = 2$, $\rho_0 = 8$ and the fixture generation rule fixed. Rebuilt source variants retain their wrapper and extraction differences; resource bundles include affinity and memory limits. A common runner records monitor timestamps at a nominal 10 ms cadence; monitoring overhead has not been shown to be zero. Extra record-field coefficient validation occurs outside task timing. These are bundle comparisons, not isolated changes to one operation.

The sensitivity and capacity designs were fixed before formal timing. Each cell has three successive sessions, with three warmup and ten measured pairs per session, balanced five/five method order and interleaved cells. Cells do not execute concurrently, and pilot observations are excluded. A failed or missing warmup withholds that cell/session’s latency summary; stopping

Rows: N / record length ℓ (bits)

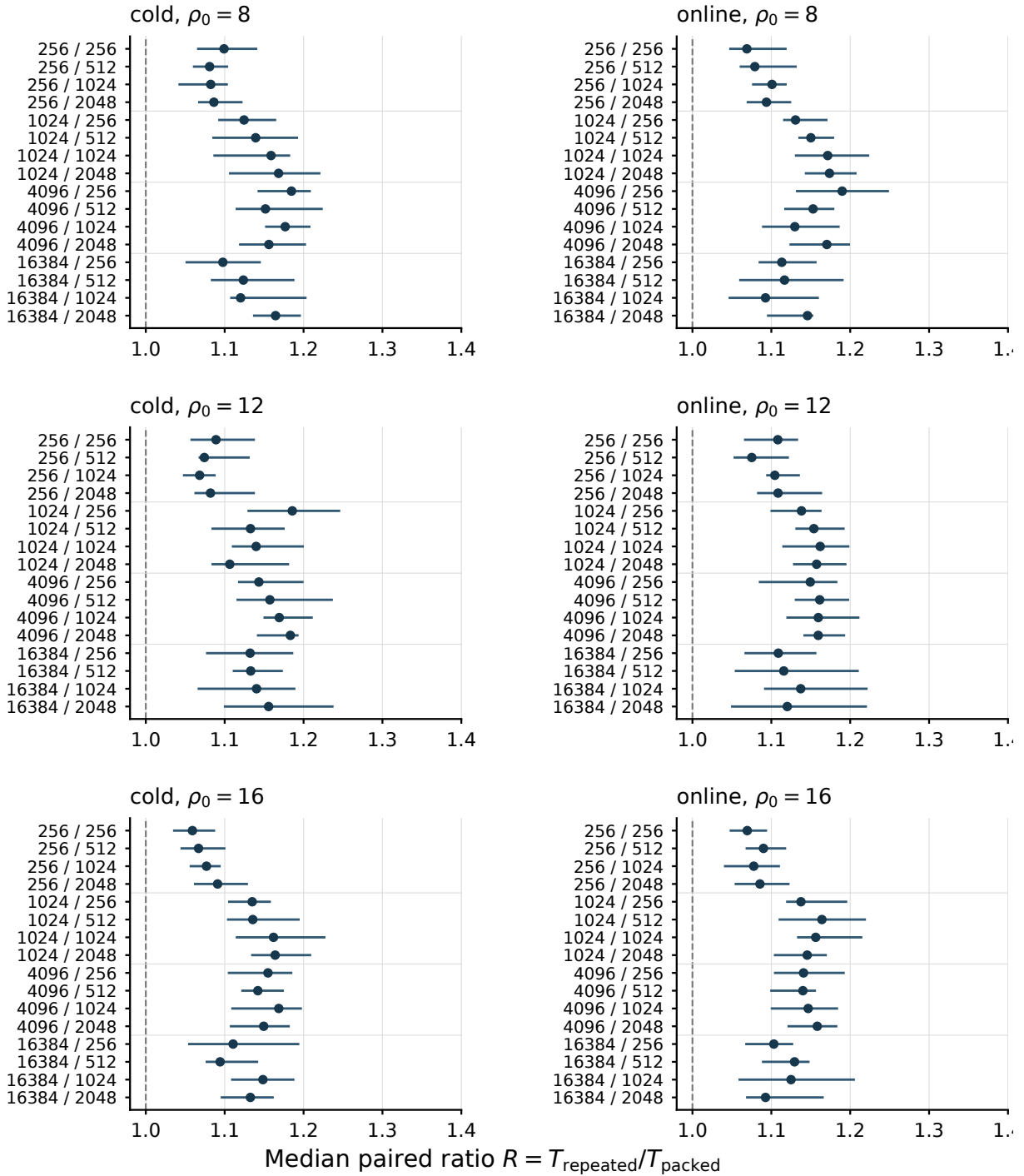


Figure 2: Primary two-record comparison: all 96 cells. Points are median paired repeated-to-packed task latency ratios; bars are pointwise 95% paired-bootstrap intervals. Cells remain separate; the figure makes no family-wise or across-grid inference.

Table 6: Multiplicity experiment at $\rho_0 = 8$, $w = 8\alpha$ and $L = 1$. Each cell contains ten measured pairs. Entries are median paired ratios with pointwise 95% intervals.

N	ℓ (bits)	View	$\alpha = 2$	$\alpha = 3$	$\alpha = 4$
1024	512	cold	1.050 [1.005, 1.078]	1.098 [1.038, 1.129]	1.140 [1.119, 1.184]
1024	512	online	1.055 [1.030, 1.082]	1.092 [1.077, 1.107]	1.151 [1.106, 1.235]
1024	2048	cold	1.036 [1.006, 1.056]	1.114 [1.078, 1.137]	1.154 [1.117, 1.186]
1024	2048	online	1.033 [0.980, 1.067]	1.096 [1.074, 1.130]	1.121 [1.088, 1.143]
4096	512	cold	1.025 [1.000, 1.052]	1.093 [1.039, 1.106]	1.122 [1.095, 1.144]
4096	512	online	1.024 [0.998, 1.039]	1.065 [1.043, 1.083]	1.108 [1.094, 1.138]
4096	2048	cold	1.013 [0.986, 1.026]	1.071 [1.061, 1.080]	1.129 [1.110, 1.152]
4096	2048	online	1.015 [0.993, 1.034]	1.066 [1.053, 1.078]	1.117 [1.097, 1.158]

The two-record entries are internal references for this experiment. Packed execution is serial; repeated tasks use α concurrent workers. This is not a fixed-total-core comparison.

rules retain failed, incomplete and unstarted tasks and forbid effect-dependent sample changes. All 702 scheduled pairs completed, including 540 measured pairs. The reported intervals for these experiments are observed ranges of three session medians, not confidence intervals or independent-day replication.

Sensitivity median ratios range from 1.123 to 1.236. In each view and implementation bundle, the descriptive median is larger under the two-CPU resource allowance. The implementation ordering changes with resource bundle and view. The two implementations' session ranges overlap within each fixed resource bundle and view. The supplemental table preserves all eight cells. These observations do not uniquely explain the primary/multiplicity difference.

Two distinct capacity boundaries.. Let $u = \alpha J/n$. For $L = 1$, a coefficient-position representation of the same fixed segments is

$$PM_{\text{position}}(X) = \sum_{\nu=1}^{\alpha} X^{(\nu-1)J} P_{i_\nu}(X).$$

It fits without overlapping coefficient ranges when $\alpha J \leq n$. Radix packing uses coefficient values instead, subject to $t \geq B^\alpha$, no-wrap and native constraints. The quantity $\lceil \alpha J/n \rceil$ is a coefficient-slot lower bound, not the reply count or runtime of an implemented alternative protocol. Other encodings and hybrid schemes are not ruled out.

The boundary $u = 1$ differs from $J = n$, where an individual record starts to require more blocks. The capacity experiment fixes $N = 1024$, $\alpha = 4$, $\rho_0 = 8$, $w = 32$, full-block extraction and the four-CPU parent allowance. Four lengths cross $u = 1$ while retaining $L = 1$; a fifth has $L = 2$ (Table 7). All ten median ratios exceed one, ranging from 1.258 to 1.339, without monotone ordering in record length. These results establish finite capacity coverage, not the speed of a coefficient-position comparator.

Table 7: Capacity experiment: $N = 1024$, $\alpha = 4$, $\rho_0 = 8$, $w = 32$, full-block extraction and the four-CPU parent allowance. Entries are median paired ratios with observed ranges of three session medians.

ℓ (bits)	J	L	u	Cold	Online
4096	512	1	0.5	1.339 [1.312, 1.376]	1.308 [1.232, 1.360]
8192	1024	1	1	1.311 [1.284, 1.318]	1.271 [1.268, 1.286]
12288	1536	1	1.5	1.311 [1.277, 1.336]	1.287 [1.287, 1.299]
16384	2048	1	2	1.319 [1.293, 1.337]	1.258 [1.257, 1.266]
65536	8192	2	8	1.294 [1.275, 1.310]	1.305 [1.293, 1.310]

Each cell has 30 measured pairs. Brackets are descriptive session ranges, not confidence intervals. Query buffers are 128/512 MiB for packed/repeated; reply buffers are 128/512 KiB for $L = 1$ and 256/1024 KiB for $L = 2$.

Table 8: Exact native buffer accounting for representative byte-aligned workloads with $\rho_0 = 8$. P/R denotes packed/repeated retrieval. Raw database size counts each record once.

N	ℓ (bits)	α	Raw (KiB)	Query P/R (MiB)	Reply P/R (KiB)
256	256	2	8	32/64	128/256
16384	2048	2	4096	2048/4096	128/256
1024	8192	4	1024	128/512	128/512
1024	16384	4	2048	128/512	128/512
1024	65536	4	8192	128/512	256/1024

One ciphertext occupies 128 KiB. These are in-memory buffer sizes; transport, framing and metadata are unmeasured.

6.6. Payload and Resource Accounting

For a declared ciphertext representation of size $|C|$,

$$|Q| = N|C|, \quad |\text{Resp}| = L|C|. \quad (17)$$

One native ciphertext occupies 131,072 bytes: two components, two RNS limbs and 4096 64-bit words. Repeated retrieval copies αN query ciphertexts and αL reply ciphertexts. These native buffers are exactly α times their packed counterparts, but transport, framing, metadata and key distribution are unmeasured. Canonical mathematical serialization can have a different size.

For byte-aligned records the raw database is $N\ell/8$ bytes, counted once regardless of multiplicity. The packed query remains larger than this database at every reported length (Table 8); even at $\ell = 65536$ its ratio to raw size is 16. Thus same-profile buffer savings do not establish raw-download or wire-traffic superiority.

Aggregate CPU work sums worker user-plus-system time, excluding parent CPU. In the multiplicity experiment, repeated-to-packed ratios of separate CPU medians are 1.972–2.052, 3.033–3.149 and 4.085–4.295 for $\alpha = 2, 3, 4$. They describe work, not latency or a fitted scaling law. Repeated RSS sums worker lifetime peaks, an upper bound rather than a synchronized peak; online staging remains included and parent RSS is separate. Phase sums likewise do not identify critical-path contributions. Full summaries and machine-readable mappings accompany the supplement.

7. Discussion and Limitations

Formal and native scope. The formal result separates conditional recovery from single-challenge index privacy. The latter assumes distribution-specific uniform-secret PLWE hardness and efficient sampling; it neither follows from empirical correctness nor requires a successful decryption event. The rounded Gaussian width remains symbolic. Its exact efficient machine realization, approximation

and computational randomness require independent evidence. The native bounded, biased sampler and OS-seeded PRNG have not been bridged to that law, and a public native noise parameter is not a proved absolute bound on integer coefficients. Consequently neither finite native passes nor the parametric bound supplies a native negligible-failure claim or a contemporary concrete-security certificate.

After a suitable joint-law and implementation bridge, task accounting could take the form $\epsilon_{\text{task}} \leq \epsilon_{\text{dec}} + d_{\text{joint}} + \epsilon_{\text{other}}$, with separately justified sampling, abort and runtime terms. An abort included in the joint-law distance is not counted again. PRNG replacement needs a separate computational failure-test bound. No such bound is established here. Independent repeated requests must also share one complete-task reliability budget, as detailed in the supplement. The privacy game gives neither symmetric PIR nor protection against malicious replies, adaptive key reuse or decryption oracles.

Empirical generality.. The measurements cover one flat backend path, one WSL2 host and finite workloads. The comparator uses concurrent private workers under the same profile within each experiment; it is not a globally optimized XPIR service with shared persistent preprocessing. External PIR timings are not aligned measurements. The capacity study adds one two-block length, and bundle sensitivity covers only one workload. Neither supports arbitrary-block, cross-host or bare-metal extrapolation.

The primary and multiplicity resource limits were enforced by their runners, but individual monitoring samples were not retained. Their transient virtual-memory and swap behavior cannot be reconstructed from the surviving endpoints and summaries. The sensitivity/capacity data retain actual monitor timestamps, yet sampled maxima are still not instantaneous peaks. This difference limits resource diagnosis without establishing invalidity of the recorded task times. Bootstrap intervals are pointwise; three successive sessions give descriptive variation. The unexplained two-record effect difference remains part of the evidence.

Design tradeoff.. Radix packing places several bounded values in each coefficient without bucket placement or additional SIMD rotations, while consuming plaintext capacity and no-wrap margin. It retains a linear-size query and does not claim general communication efficiency. A deployment would require a joint choice of correctness and security parameters, a justified sampler and randomness implementation, transport measurements, and comparisons with optimized preprocessing and query-compression alternatives.

8. Conclusion

Radix-packed retrieval combines bounded segments from several records within each reply coefficient while retaining an XPIR-style linear interface. Its specification makes encoding, exact capacity, integer no-wrap correctness and single-challenge privacy explicit. The privacy reduction uses $N + 1$ uniform-secret PLWE samples and the stated unit-probability loss; the full-task reliability bound remains parametric in the specified rounded Gaussian law.

On the measured WSL2 host, the primary two-record comparison finds lower median completed-task latency than concurrent same-profile repeated XPIR throughout the tested grid. Separate multiplicity, sensitivity and capacity experiments describe additional finite regimes, including one two-block workload. Native buffer savings remain relative to repeated execution; the flat query still exceeds raw database size at these lengths. The unresolved native/formal bridge, distinct experimental resource contracts and unmeasured transport delimit the conclusions.

Artifact availability

The archived reproduction package provides manuscript sources, observations, frozen analysis, presentation code and build/functional validation instructions. It preserves the separate experimental contracts and their source-to-result mappings.

References

- [1] B. Chor, E. Kushilevitz, O. Goldreich, M. Sudan, Private information retrieval, *Journal of the ACM* 45 (6) (1998) 965–981. doi:10.1145/293347.293350.
- [2] E. Kushilevitz, R. Ostrovsky, Replication is not needed: Single database, computationally-private information retrieval, in: *Proceedings of the 38th Annual Symposium on Foundations of Computer Science, IEEE, 1997*, pp. 364–373. doi:10.1109/SFCS.1997.646125.
- [3] C. Aguilar-Melchor, J. Barrier, L. Fousse, M.-O. Killijian, XPIR: Private information retrieval for everyone, *Proceedings on Privacy Enhancing Technologies* 2016 (2) (2016) 155–174. doi:10.1515/popets-2016-0010.
URL <https://petsymposium.org/popets/2016/popets-2016-0010.php>
- [4] H.-C. Hsu, Z.-Y. Liu, R. Tso, K. Chen, Multi-value private information retrieval using homomorphic encryption, in: *2020 15th Asia Joint Conference on Information Security (AsiaJCIS), IEEE, 2020*, pp. 82–88. doi:10.1109/AsiaJCIS50894.2020.00024.
- [5] XPIR Team, XPIR: Private information retrieval for everyone (software), <https://github.com/XPIR-team/XPIR>, source revision identified in the reproduction package.
- [6] S. Angel, H. Chen, K. Laine, S. Setty, PIR with compressed queries and amortized query processing, in: *2018 IEEE Symposium on Security and Privacy (SP), IEEE, 2018*, pp. 962–979. doi:10.1109/SP.2018.00062.
URL <https://eprint.iacr.org/2017/1142>
- [7] M. H. Mughees, L. Ren, Vectorized batch private information retrieval, in: *2023 IEEE Symposium on Security and Privacy (SP), IEEE, 2023*, pp. 437–452. doi:10.1109/SP46215.2023.10179329.
URL <https://eprint.iacr.org/2022/1262>
- [8] J. Liu, J. Li, D. Wu, K. Ren, PIRANA: Faster multi-query PIR via constant-weight codes, in: *2024 IEEE Symposium on Security and Privacy (SP), IEEE, 2024*, pp. 4315–4330. doi:10.1109/SP54263.2024.00039.
URL <https://eprint.iacr.org/2022/1401>
- [9] A. Henzinger, M. M. Hong, H. Corrigan-Gibbs, S. Meiklejohn, V. Vaikuntanathan, One server for the price of two: Simple and fast single-server private information retrieval, in: *32nd USENIX Security Symposium (USENIX Security 23), USENIX Association, 2023*, pp. 3889–3905.
URL <https://www.usenix.org/conference/usenixsecurity23/presentation/henzinger>
- [10] A. Burton, S. J. Menon, D. J. Wu, Respire: High-rate PIR for databases with small records, in: *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security, ACM, 2024*, pp. 1463–1477. doi:10.1145/3658644.3690328.
URL <https://eprint.iacr.org/2024/1165>
- [11] Z. Zhang, R. Mao, Z. Zhao, H. Pan, Y. Fu, Y. Liu, D. Li, Y. Chen, B. Zhang, C. Ren, J. Dong, Z. Guan, S. Bian, SPIRIT: Batch hintless single-server PIR via stateful ciphertext conversion, in: *35th USENIX Security Symposium (USENIX Security 26), USENIX Association, 2026*, pp. 4523–4542.
URL <https://www.usenix.org/conference/usenixsecurity26/presentation/zhang-zhou>

- [12] Z. Brakerski, V. Vaikuntanathan, Fully homomorphic encryption from Ring-LWE and security for key dependent messages, in: *Advances in Cryptology – CRYPTO 2011*, Springer, 2011, pp. 505–524. doi:10.1007/978-3-642-22792-9_29.
- [13] V. Lyubashevsky, C. Peikert, O. Regev, A toolkit for Ring-LWE cryptography, *Cryptology ePrint Archive*, Report 2013/293, author version dated May 16, 2013; Lemma 2.24 and Claim 2.25 (2013).
URL <https://eprint.iacr.org/2013/293>

Appendix A. Supporting Proofs

Appendix A.1. Rounded-noise tails

Proof. The rounding cells for $-h, \dots, h$ fill $[-h - 1/2, h + 1/2)$, apart from boundary conventions. Hence $|Z| > h$ iff $|Y| \geq h + 1/2$, up to half-integer ties, which have probability zero under the continuous Gaussian. Integrating its two symmetric tails proves (9). The non-strict identity follows from integer-valuedness: $|Z| \geq h$ iff $|Z| > h - 1$. $\square$

Appendix A.2. Block-support noise bound

Proof. There are m_j valid segment positions in block j , each in $[0, B - 1]$, and all other coefficients are zero. Since $(L - 1)n < J \leq Ln$, actual blocks are nonempty. Unused high bits in the last segment only reduce the stated bound. For degree- $< n$ representatives, the coefficient of a negacyclic product is

$$[X^k](Pe) = \sum_{u+v=k} P_u e_v - \sum_{u+v=k+n} P_u e_v.$$

The minus sign is exactly the reduction $X^n = -1$. For each u there is one $v = (k - u) \bmod n$ in these two sums. The triangle inequality therefore bounds its absolute value by $\sum_u |P_u| \|e\|_\infty$. On G_h , each $\|e_i\|_\infty \leq h$; summing $\|\hat{P}_{i,j} e_i\|_\infty \leq m_j(B - 1)h$ over all N records proves (12). $\square$

Appendix A.3. Normal-form specialization

Specialization sketch. Given $(a_i, b_i)_{i=0}^\tau$, output zero if a_0 is not a unit. Otherwise give $\mathcal{D}$ the pairs

$$A_i = -a_0^{-1} a_i, \quad B_i = b_i + A_i b_0, \quad i \in [\tau].$$

In the real case, $B_i = A_i e_0 + e_i$, so the new shared secret is $e_0 \leftarrow \chi$ and all remaining errors still have law χ . The unit event depends only on a_0 , hence does not bias e_0 ; unit multiplication preserves the independent uniform A_i . In the uniform case, conditional on a_0, b_0 and the unit event, the transformed pairs are independent uniform pairs. Thus $\mathcal{B}$'s gap is exactly p_{unit} times $\mathcal{D}$'s gap. Unit testing and inversion are polynomial-time operations: use the multiplication matrix of a_0 over $\mathbb{Z}_q$, test whether its determinant is coprime to q , and invert when it is. No factorization of q is required. $\square$

Appendix A.4. Ciphertext scaling and translation

Proof. For each normal-form pair $(A, B = As + E)$ and canonical message lift $\hat{m}$, set

$$(a, b) = (tA, tB + \hat{m}).$$

Then $b = as + tE + \hat{m}$, exactly XPIR's SKE syntax, or $C = (b, -a)$ in our ordering. In the uniform branch, $(tA, tU + \hat{m})$ remains uniform because t is a unit modulo q . Apply the same bijection to all samples sharing s . This is the noise-scaling observation of [12, Proposition 1], followed by message translation; the argument uses only $\gcd(t, q) = 1$, including for composite q . $\square$

Appendix A.5. Single-challenge privacy reduction

Proof. First construct a normal-form distinguisher $\mathcal{D}$. Given pp and N normal-form-or-uniform pairs (A_i, B_i) , it runs $\mathcal{A}_1(pp)$ to obtain (M, I_0, I_1, σ) . If the inputs are illegal, it outputs a fair bit, matching the experiment's convention. Otherwise it privately chooses $\beta \leftarrow \{0, 1\}$ and forms

$$C_i = (tB_i + v_{I_\beta, i}, -tA_i), \quad Q = (C_1, \dots, C_N). \quad (\text{A.1})$$

The constants $v_{I_\beta, i}$ use their canonical message lifts. It computes $P = \text{DBEncode}(M)$ and $\text{Resp} = \text{ReplyGen}(pp, P, Q)$, gives (Q, Resp) to $\mathcal{A}_2$ with precisely the other inputs of the experiment, and outputs one iff the guess β' equals β . It needs neither s nor an encryption or decryption oracle.

For real normal-form samples, $B_i = A_i s + E_i$, with one shared $s \leftarrow \chi$ and fresh independent $E_i \leftarrow \chi$. By Corollary 3, setting $a_i = tA_i$ gives

$$C_i = (a_i s + tE_i + v_{I_\beta, i}, -a_i),$$

exactly a fresh $\text{Enc}_s(v_{I_\beta, i})$. This preserves the entire joint law of all N query ciphertexts, including their shared secret, not just individual marginals. The first-stage output has the same law as in the game because it depends only on pp and adversary randomness, independently of the samples and of β . The encoded database and reply use the actual deterministic protocol algorithms. Hence (Q, Resp) has exactly the formal protocol view's distribution, and

$$\Pr[\mathcal{D}(pp, \mathcal{F}_{\text{real}}^{(N)}) = 1] = \Pr[\text{privacy experiment succeeds}].$$

For uniform samples, A_i, B_i are independent uniform ring elements. Multiplication by the unit t , negation, and translation by the fixed $v_{I_\beta, i}$ preserve uniformity. Conditional on every legal first-stage output and either value of β , the entire query is uniform in $(R_q^2)^N$, independently of β . The reply is deterministic PPT post-processing of this query and the fixed database. Therefore (Q, Resp) remains independent of β . The same is true of the adversary's resulting guess. Accounting also for the independent fair output on illegal input,

$$\Pr[\mathcal{D}(pp, \mathcal{F}_{\text{unif}}^{(N)}) = 1] = \frac{1}{2}.$$

Under the success-minus-1/2 convention of the privacy game and the acceptance-gap convention for PLWE, subtraction yields

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{priv}}(\kappa) = \text{Adv}_{\mathcal{D}}^{\text{NF}}(\kappa; N).$$

Compose $\mathcal{D}$ with Corollary 2 at $\tau = N$ to obtain $\mathcal{B}$. It consumes $N + 1$ uniform-secret ring samples, uses the first for normal form, and passes the remaining N transformed pairs to $\mathcal{D}$. Conditional on the unit event its simulation is the one just analyzed. With output zero on abort, its uniform-branch acceptance probability is $p_{\text{unit}}/2$, not $1/2$; both branches are scaled by the same p_{unit} . Thus

$$\text{Adv}_{\mathcal{B}}^{\text{PLWE}}(\kappa; N + 1) = p_{\text{unit}} \text{Adv}_{\mathcal{D}}^{\text{NF}}(\kappa; N),$$

which gives the stated bound. There is no extra factor 2, N , or 2α : the reduction simulates the joint query in one step.

All steps are polynomial-time: the normal-form operations are as in Corollary 2, both adversary stages are PPT, and encoding and reply generation operate on a polynomial-size database and query using polynomially many ring operations with polynomial bit complexity. Deterministic reply generation adds no distinguishing loss. $\square$